C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Reactivity and Extraction **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Reactivity series (most reactive first)				K, Na, Li, Ca, Mg, C, Zn, Fe, H, Cu
2	Products of a metal with water				Metal hydroxide + hydrogen
3	Products of a metal with acid				Salt + hydrogen
4	Oxidation (in terms of oxygen)				Gain of oxygen
5	Reduction (in terms of oxygen)				Loss of oxygen
6	Metals extracted by reduction with carbon				Those less reactive than carbon, e.g. iron
7	Metals extracted by electrolysis				Those more reactive than carbon, e.g. aluminium
8	Metal found uncombined in the Earth				Gold
9	Displacement reaction				More reactive metal displaces a less reactive one from a compound
10	Ore				Rock containing enough metal to extract

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 2: Redox and Salts **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Oxidation (in terms of electrons)				Loss of electrons
2	Reduction (in terms of electrons)				Gain of electrons
3	Spectator ion				Ion that does not change in the reaction
4	Acid + alkali				Salt + water
5	Acid + metal oxide				Salt + water
6	Acid + metal carbonate				Salt + water + carbon dioxide
7	Salts made from hydrochloric acid				Chlorides
8	Salts made from sulfuric acid				Sulfates
9	Salts made from nitric acid				Nitrates
10	Base				Substance that neutralises an acid

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 3: pH and Strong and Weak Acids **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	pH of a neutral solution				7
2	pH of acids				Below 7
3	pH of alkalis				Above 7
4	Ion that makes solutions acidic				H^+
5	Ion that makes solutions alkaline				OH^-
6	Ionic equation for neutralisation				$H^+ + OH^- \rightarrow H_2O$
7	Strong acid				Completely ionised in water
8	Weak acid				Partially ionised in water
9	Examples of strong acids				Hydrochloric, nitric, sulfuric
10	Change in $[H^+]$ when pH falls by 1				× 10

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 4: Electrolysis **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Electrolysis				Splitting a compound using electricity
2	Electrolyte				Molten or dissolved ionic compound
3	Cathode				Negative electrode
4	Anode				Positive electrode
5	Ions attracted to the cathode				Positive ions
6	Products of electrolysis of molten lead bromide				Lead and bromine
7	Product at the cathode in aqueous solution				Hydrogen, unless the metal is less reactive than hydrogen
8	Product at the anode in aqueous solution				Oxygen, unless halide ions are present
9	Products of electrolysis of brine				Hydrogen, chlorine, sodium hydroxide
10	Half-equation for chloride ions at the anode				$2Cl^- \rightarrow Cl_2 + 2e^-$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 5: Electrolysis of Aluminium Oxide **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Aluminium is extracted from				Aluminium oxide
2	What aluminium oxide is dissolved in				Molten cryolite
3	Why cryolite is used				Lowers the melting point, saving energy
4	Material of the electrodes				Carbon (graphite)
5	Product at the cathode				Aluminium
6	Product at the anode				Oxygen
7	Why the anodes must be replaced				Carbon reacts with oxygen to form CO_2
8	Half-equation at the cathode				$Al^{3+} + 3e^- \rightarrow Al$
9	Half-equation for hydrogen at the cathode				$2H^+ + 2e^- \rightarrow H_2$
10	Half-equation for oxygen at the anode				$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C4 Chemical Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 6: Titrations Paper 1 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Purpose of a titration				Find volumes of acid and alkali that exactly react
2	Indicators for a titration				Phenolphthalein or methyl orange
3	Why universal indicator is not used				Gradual colour change
4	Phenolphthalein colours: acid / alkali				Colourless / pink
5	Methyl orange colours: acid / alkali				Red / yellow
6	Why a white tile is used				To see the colour change clearly
7	How acid is added near the end point				Dropwise
8	Where the burette is read				Bottom of the meniscus
9	Concordant results				Within 0.10 cm^3 of each other
10	Burette is rinsed with				The acid it will contain

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.